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Received: 30 April 2022 | Revised: 23 September 2022 | Accepted: 5 October 2022

DOI: 10.1002/rob.22127

RESEARCH ARTICLE

WILEY

Digital twin-assisted fault diagnosis system for robot joints with insufficient data

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Funding information
Liaoning Revitalization Talents Program (No. XLYC2007031); High-level innovation and entrepreneurship team (No. XLYC2008016); Natural Science Foundation of Liaoning (No. 2021-KF-12-09); National Science Foundation of China (No. 52275119); National Science Foundation of China (No. 52075348)

Abstract

The robot joint is an important component of the construction robot, and its fault diagnosis can ensure the exact execution of building jobs, stable operation, and timely prevention of probable safety mishaps. However, deep learning-based fault diagnosis needs a multitude of measured fault data, which is difficult to obtain for various reasons. To solve the problem of insufficient data, a digital twin-assisted fault diagnosis system for robot joints is proposed. First, a simplified dynamics model of the robot joint is developed to generate the virtual entity data which can be used as the X-domain data for the digital twin model. Second, a CycleGAN-based digital twin model is proposed to map the virtual entity (X-domain) data to the physical entity (Y-domain) utilizing only a small amount of measured data. In the end, a test-rig for the robot joint is built to simulate the robot's working conditions, and the CNN-ResNet classifier is utilized to verify the effectiveness of the simulated data generated by the digital twin model. The results show that the fault diagnosis accuracy can be increased from 32.5% to 98.86% utilizing only 400 sets of measured data.

KEYWORDS
construction robot, CycleGAN, digital twin model, fault diagnosis, insufficient data, robot joint, RV reducer

1 | INTRODUCTION

The construction robot is an intelligent construction equipment that can improve the working environment and increase efficiency. Considering the building process in the construction industry is complex, there have been a multitude of intelligent construction robots that assist in construction, such as wall construction robots (Han et al., 2006), painting robots (Asadi et al., 2018), and maintenance building robots (Follini et al., 2020). Construction robots can improve construction efficiency, ensure the health and safety of construction workers, and solve the labor shortage in the construction industry (Mascaro et al., 2021).

In recent decades, robotics has begun to penetrate the construction industry. In 1982, the SSR-type robot for spraying refractory covering material on steel skeletons was developed by Shimizu Construction Company in Japan, just as shown in Figure 1a. As shown in Figure 1b, the Semi-Automated Mason is a robot that has been developed by the Construction Robotics company in New York, and it is several times more efficient than a human being. As shown in Figure 1c, the robot, model Robo Tab-2000B, had 6 DOF to achieve translation and rotation of gypsum board in all directions. In 2014, the floor tile-laying robot called Mobile Robotic Triling (MRT) was developed by Future Cities Lab in Singapore in conjunction with ETH Zurich, just as shown in Figure 1d. At present, traditional manual building construction technology has a slew of issues, such as low efficiency, high labor intensity, and high cost, which have become shortcomings of the new intelligent building construction technology. Therefore, only by developing construction robots can the change in building construction be realized.

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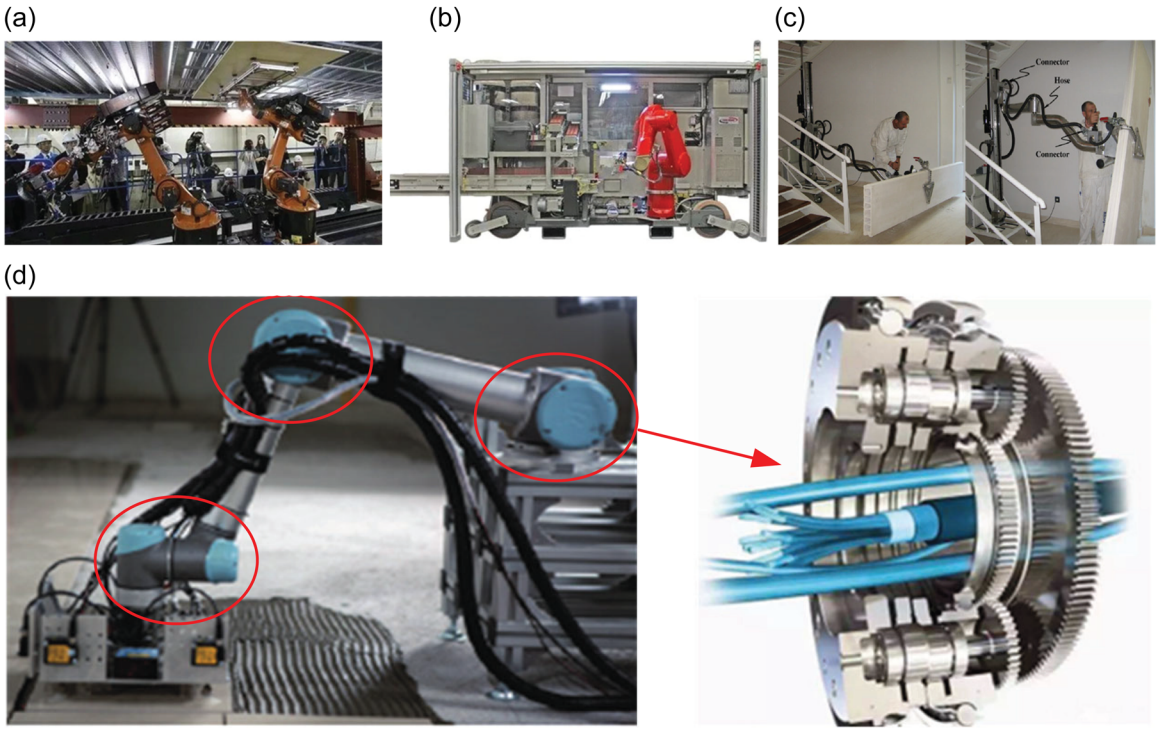


FIGURE 1 Construction robots: (a) SSR type robot for spraying refractory covering material on steel skeleton, (b) Semi-Automated Mason, (c) RoboTab-2000B, and (d) Mobile Robotic Triling

The RV reducer is widely used as the construction robot joint because of its high transmission efficiency, high torque, and smooth transmission performance (Chen, 2018). However, it is easily damaged under long-term work and overload operation conditions. At present, there have been a multitude of scholars pouring attention into transmission efficiency (Wang, 2021), optimization of design (Yang et al., 2021) and error analysis of RV reducers (Li et al., 2018), but little into the fault diagnosis aspect. The existing fault diagnosis of RV reducers is mostly based on deep learning (Chen et al., 2021; Long et al., 2021; Peng & Wang, 2019). Traditional fault characteristics are extracted from vibration signals (Shi & Shang, 2022), sound signals (Bai et al., 2022) or sound-vibration signals (Shi et al., 2022) to achieve accurate fault classification. Considering the complex construction environment and working conditions of construction robots, it does not seem feasible to collect sound signals with the interference of a strong noise background. Therefore, deep learning based on vibration signals has become the mainstream method for fault diagnosis of robots. A sparse deep learning method was proposed by (Sun et al., 2018) to reduce the risk of overfitting in deep networks with a large number of layers and neurons. A deep transfer learning-based fault diagnosis framework was proposed by Shao et al. (2019a) to obtain faster model convergence and higher fault diagnosis accuracy. A one-dimensional residual convolutional autoencoder-based feature learning was proposed by Yu and Zhou (2020) to better extract the features of vibration signals under noisy conditions. The above deep learning-based fault diagnosis methods all require sufficient measured data to train the neural network. But the measured data on faulty robot joints are difficult to obtain under the actual working conditions. The reasons may be as follows: (1) most of the construction robots are in normal working condition; (2) obtaining failure data through experiments is a time-consuming and costly process; (3) some fault data cannot even be obtained through experiments.

To provide different kinds of fault data for the RV reducer, some scholars have established dynamics models (Qiao et al., 2020; Wang et al., 2019; Xie et al., 2022; Xu et al., 2019). However, there are such varying degrees of idealization in the dynamics modeling that the dynamics model cannot completely describe the motion behavior of the internal components of RV reducers. This will cause there to be a gap between the simulated signals of the dynamics model and the measured signals of the RV reducer. Therefore, the key to improving the accuracy of the fault diagnosis of the RV reducer is how to reduce the gaps between the simulated dynamic response and the measured dynamic response. To establish a relationship between simulated dynamic response and measured dynamic response, digital twin provides a feasible method. The concept of the digital twin can date back to 2003, when Professor Grieves developed the Mirror Space Model in a product life cycle management class at the University of Michigan (Grieves, 2005). Digital twin is a virtual representation of the physical entity that is created digitally. It can also develop or grow new abilities for the physical entity utilizing a variety of methods, including utilizing data to imitate the physical entity's behavior in the real world, decision iterative optimization, and so on

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Tao et al. (2019). Nowadays, digital twin has application prospects in intelligent manufacturing (Leng et al., 2021), complex electro-mechanical equipment operation and maintenance (Tao et al., 2018), aerospace (Liu et al., 2021), and other fields. Nowadays, a multitude of scholars have already applied digital twin to fault diagnosis. A digital twin model of the autoclave was established by Wang et al. (2021). And it can be used to enhance the fault prediction of the autoclave by generating simulated data, which is used to replace scarce data from actual failures. A construction method for digital twin was proposed by Karve et al. (2020), which can be used for damage detection and prediction of uncertain crack propagation. A framework of mechanical fault diagnosis was proposed by Xia et al. (2021), which consisted of digital twin and deep transfer learning technologies. And the limitations or unavailableness of mechanical equipment fault state data could be solved under variable working conditions or variable system characteristics. A digital twin system for structural damage detection was designed by Ritto and Rochinha (2021), and was based on models and machine learning technologies to provide intelligent decision-making solutions. These works have contributed to the development of digital twin by establishing a number of effective digital twin models in a variety of fields. However, few studies pay much attention to the mapping of digital twin. There are also some digital twin works in robots. A method based on physical models and digital twin was proposed by Aivaliotis et al. (2019) to calculate the remaining useful life of mechanical equipment. An open-source 3D printed robotic arm was developed by Matulis and Harvey (2021), and its digital twin model utilizing Unity was also established. The digital twin based on multi-sensor data fusion technology was combined with intelligent inspection robots by He et al. (2021) to explore the sustainability of massage chairs. Despite the fact that all of the works mentioned above developed effective digital twin models for the robotics field, only a few of them focused on how the digital twin can be used for fault diagnosis.

In view of the above problems, a digital twin-assisted fault diagnosis system for robot joints is proposed in this work, which can figure out the problem of insufficient data. The main work and innovations of this paper are as follows. (1) The virtual entity of the digital twin model is based on the dynamics model of gears and can generate gear vibration signals without noises and faults. (2) A CycleGAN-based digital twin model trained utilizing a small number of samples is proposed for generating simulated vibration signals. (3) The simulated vibration signals generated by the digital twin can effectively capture the fault characteristics and improve the accuracy of fault diagnosis. This paper is organized as follows. In Section 2, the theoretical background of the digital twin-assisted fault diagnosis system is presented. The framework and specific processes of the system are given in Section 3. In Section 4, we built a test-rig of the robot joint and evaluate the performance of the simulated vibration signals generated by the digital twin. Some comparisons, analyses, and discussions on fault diagnosis are also presented. Finally, the conclusions of this work are summarized in Section 5.

2 | THEORETICAL BACKGROUND

The digital twin-assisted fault diagnosis system consists of three key methods: (1) the simplified robot joint dynamics model; (2) the CycleGAN-based digital twin model; and (3) the deep learning-based fault diagnosis. This section will introduce the fundamentals of the methods mentioned above.

2.1 | The simplified robot joint dynamics model

The RV reducer is a closed two-stage planetary-transmission mechanism that consists of planetary gears and cycloid-pinwheels. Because of its high transmission efficiency, high torque, and smooth transmission performance, the RV reducer is widely used in the transmission joints of construction robots. Its transmission principles are shown in Figure 2.

In this paper, we only consider the failure of the planetary wheel in the primary drive of the RV reducer, so the model of the RV reducer can be simplified to a planetary wheel system. In addition, since the planetary wheel in the RV reducer engages only the sun wheel, its dynamics model can be further simplified to the meshing of two gears. It is worth noting that the gear meshing process involves both single and double tooth meshing. Therefore, the total effective mesh stiffness K_t varies with the meshing of a pair of gears. When a pair of gears is meshing with a single tooth, the K_t can be expressed as

$$K_t = 1 \Big/ \left(\frac{1}{K_h} + \frac{1}{K_{ai}} + \frac{1}{K_{bi}} + \frac{1}{K_{si}} \right) \quad (i = 1, 2), \tag{1}$$

where K_h is Hertzian contact stiffness; K_a is axial compressive stiffness; K_b is bending compressive stiffness; K_s is shear compressive stiffness. And $i = 1$ represents the active gear and $i = 2$ represents the driven gear. When a pair of gears is in double tooth meshing duration, the K_t can be expressed as

$$K_t = \sum_{i=1}^{2} 1 \Big/ \left(\frac{1}{K_h} + \frac{1}{K_{ai}} + \frac{1}{K_{bi}} + \frac{1}{K_{si}} \right). \tag{2}$$

More detailed calculation process of (1) and (2) can be found in Wu et al. (2008).

2.2 | The CycleGAN-based digital twin model

In this work, we will propose a novel digital twin model based on CycleGAN. CycleGAN was first proposed by Zhu et al. (2017) and was widely used for image style conversion. The advantage of CycleGAN is that the paired training datasets are not a must, which are difficult to obtain but may be necessary to train traditional neural networks. The CycleGAN consists of two generators (G_X, G_Y) and two discriminators (D_X, D_Y), just as shown in Figure 3. The purpose of the generator is to generate data, that is similar enough to the target

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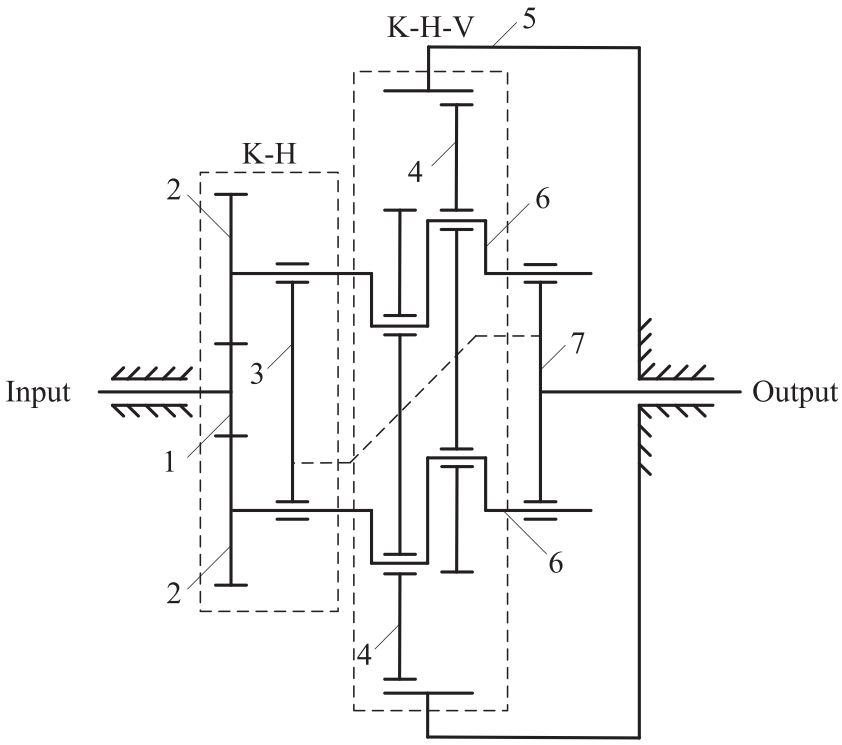


FIGURE 2 Transmission principle of the RV reducer: 1—Sun gear; 2—Planetary gear; 3—Tie bar; 4— Cycloid gear; 5—Pin teeth; 6—Crankshaft; 7—Output side

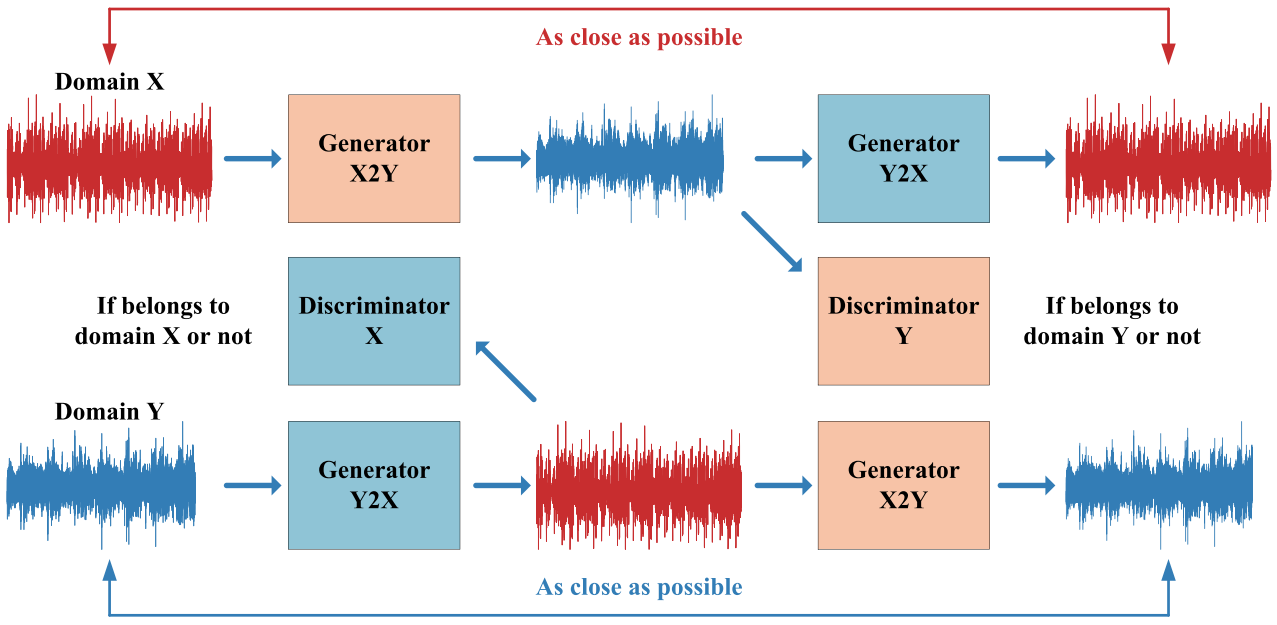


FIGURE 3 The principle of CycleGAN

domain *Y*, to deceive the discriminator. And the job of the discriminator is to do its best to recognize the data generated by the generator. The whole training process of CycleGAN is a continuous confrontation between the generator and the discriminator. As a result, the training of CycleGAN is complete when the generator and discriminator achieve equilibrium.

The reason why CycleGAN can achieve the transformation from domain X to domain Y is that it introduced a new Cyclic Consistency Loss ($L_{cyc}(G_X, G_Y)$) on top of Adversarial Loss (L_{GAN}) of the traditional GAN. The $L_{GAN}(G_X, D_Y)$ can be expressed as

$$L_{GAN}(G_X, D_Y) = E[\log D_Y(y)] + E[\log(1 - D_Y(G_X))]. \quad (3)$$

The $L_{GAN}(G_Y, D_X)$ can be defined in the same way. And the $L_{cyc}(G_X, G_Y)$ can be expressed as

$$L_{cyc}(G_X, G_Y) = E[\|G_Y(G_X) - x\|_1] + E[\|G_X(G_Y) - y\|_1]. \quad (4)$$

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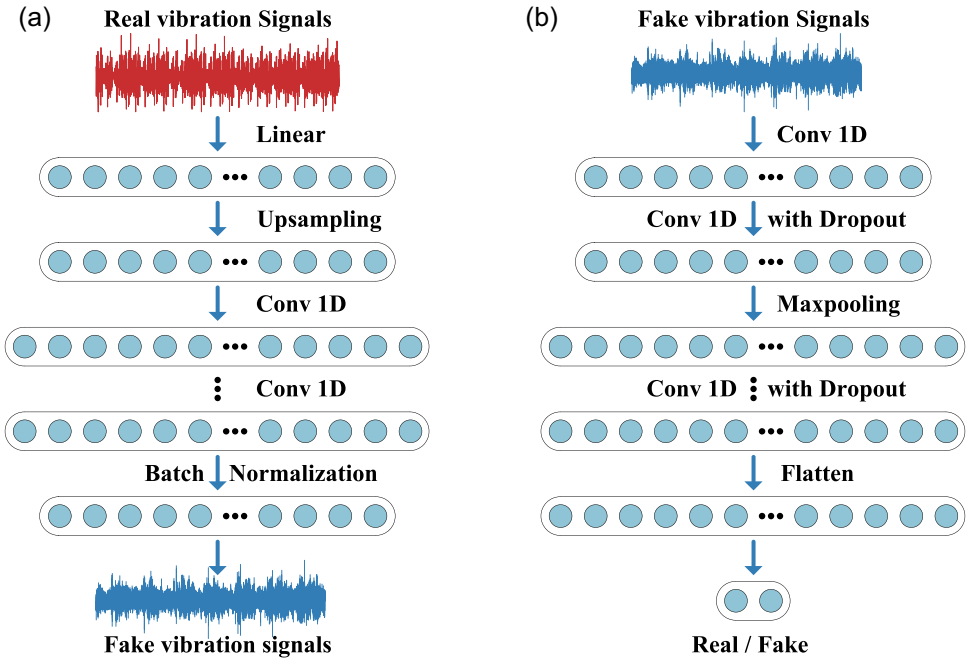


FIGURE 4 The structure of the (a) generator and (b) discriminator

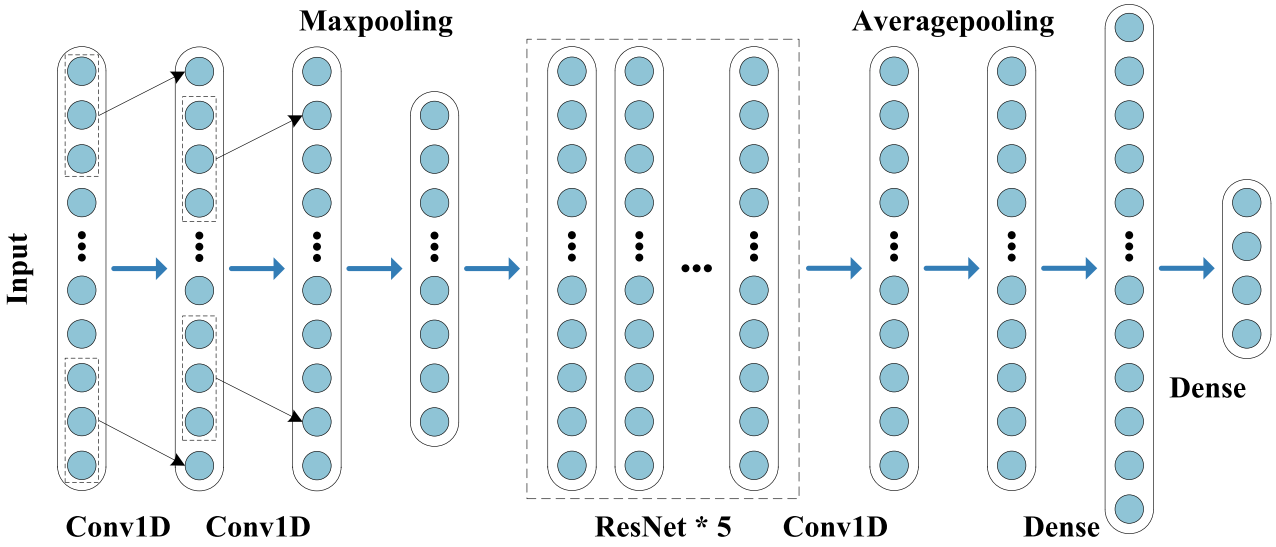


FIGURE 5 The structure of CNN-ResNet

Based on (3) and (4), the total loss of CycleGAN can be expressed as

$$L_{CycleGAN} = L_{GAN}(G_X, D_Y) + L_{GAN}(G_Y, D_X) + \lambda L_{cyc}(G_X, G_Y), \quad (5)$$

where λ is a controlling ratio. More detailed explanation of parameters in (3)–(5) can be found in (Zhu et al., 2017). With $L_{cyc}(G_X, G_Y)$, the generator generates data that not only learns the features of the target domain Y, but also retains the features of the source domain X (Qin et al., 2022). The structure of the generator and discriminator used in this paper is shown in Figure 4.

2.3 | Deep learning-based fault diagnosis

Currently, deep learning-based fault classification techniques are one of the most common means of fault diagnosis. To implement fault diagnosis of robot joints and to evaluate the effectiveness of the proposed digital twin model, a CNN-ResNet-based classifier is proposed in this work. The convolution layer, pooling layer, residual layer, and full connection layer are the main components of this classifier, which is shown in Figure 5. After training, the classifier can adequately capture the fault characteristics of the vibration signals. As a result, the classifier can directly output the fault type with the highest probability when the sensor data is input to the classifier.

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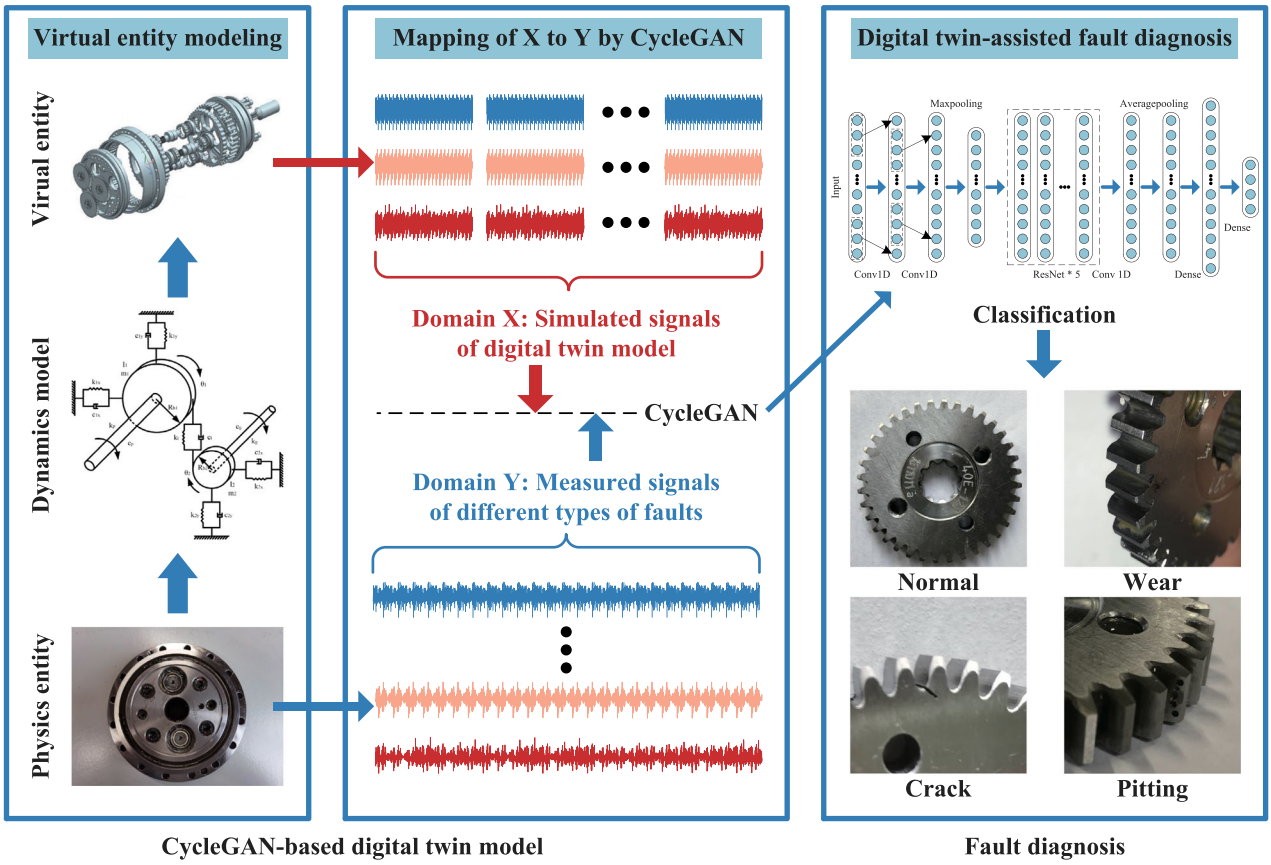


FIGURE 6 The digital twin-assisted fault diagnosis system

3 | DIGITAL TWIN-ASSISTED FAULT DIAGNOSIS SYSTEM FOR ROBOT JOINTS

In this section, the framework and the specific process of the digital twin-assisted fault diagnosis system are described. And we also summarize the advantages of the system at the end of this section.

3.1 | Framework of digital twin-assisted fault diagnosis system

The digital twin-assisted fault diagnosis framework proposed in this paper consists of three main parts: virtual entity modeling, virtual-to-physical mapping, and digital twin-assisted fault diagnosis. First, the virtual entity model is built based on the gear dynamics and generates the vibration signals as the domain X. Second, a small amount of the measured data collected by the sensor is used as the domain Y. Furthermore, CycleGAN is used to map the domain X to the domain Y, which means the virtual domain is mapped to the physical domain. The above constitutes the CycleGAN-based digital twin model. The simulated vibration signals generated by the digital twin model can effectively improve the fault diagnosis accuracy. Finally, CNN-ResNet is used for fault classification. As shown in Figure 6, the general framework of the digital twin-assisted fault diagnosis system mainly includes the following elements:

1. Virtual entity modeling: The virtual entity model based on gear dynamics can generate noiseless and faultless vibration signals, which are used as X-domain data.
2. Mapping of virtual to physical domain: The vibration signals generated by the dynamics model are mapped to the physical domain by CycleGAN. The simulated vibration signals mapped to the physical domain will have the same distribution and fault characteristics as the measured vibration signals.
3. Digital twin-assisted fault diagnosis: The data generated by the digital twin model can be used directly as training datasets. And the datasets can effectively assist in fault diagnosis and provide good accuracy.

3.2 | Virtual entity modeling

It is among the most significant forms of digital twin to create a correct model of the virtual entity. Therefore, only by building the correct model of the virtual entity can it accurately simulate the physical entity. In this paper, the 6-DOF gear system dynamics model

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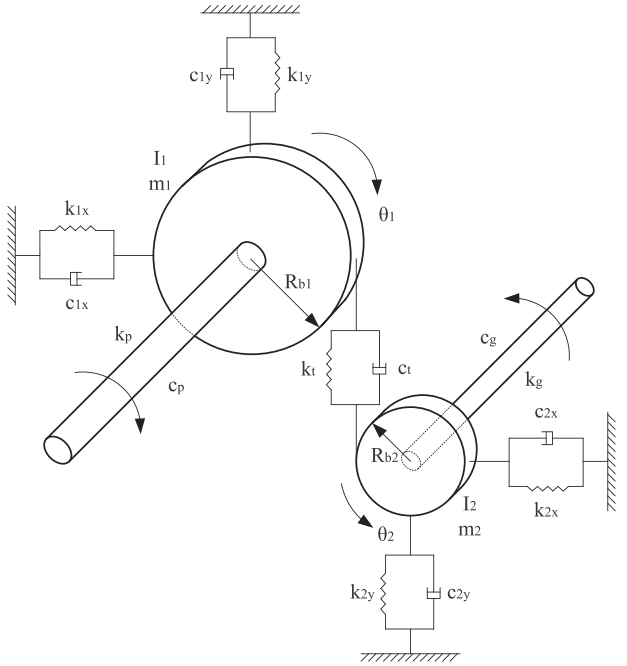


FIGURE 7 Dynamics model of gear system with 6-DOF

is used to represent the changes in the physical entity, which is shown in Figure 7.

According to Newtonian Mechanics, the dynamics equations are established as follows:

$$\begin{cases} m_1\ddot{y}_1 = F_k + F_c - F_u - F_{uc}, \\ m_2\ddot{y}_2 = F_k + F_c - F_l - F_{lc}, \\ I_1\ddot{\theta}_1 = M_{pk} + M_{pc} - R_{b1}(F_k + F_c), \\ I_2\ddot{\theta}_2 = R_{b2}(F_k + F_c) - M_{gk} - M_{gc}, \\ I_m\ddot{\theta}_m = M_1 - M_{pk} - M_{pc}, \\ I_b\ddot{\theta}_b = -M_2 + M_{gk} + M_{gc}, \end{cases} \tag{6}$$

where $\ddot{y}_1, \ddot{y}_2$ are the acceleration of the gear 1 and 2 in the vertical direction respectively; $\ddot{\theta}_1, \ddot{\theta}_2$ are the angular displacement of the gear 1 and 2 respectively; $\ddot{\theta}_m, \ddot{\theta}_b$ are the angular displacement of motor and load respectively. And more parameters in Figure 7 and (6) can be found in Wu et al. (2008). After the virtual entity model is completed, the simulation vibration signals can be generated by following: (1) Measure the dimensional parameters and working conditions of the failed planetary wheel of the RV reducer according to the actual situation; (2) Set the parameters of the dynamics model to be the same as the actual situation; (3) Initialize the parameters and solve (6) utilizing the Runge–Kutta method. And then the simulated vibration signals at any time can be obtained.

3.3 | Mapping of the virtual to the physical domain

As shown in Figure 8, the mapping process of CycleGAN mainly covers the following aspects: (1) Discrete measured data and simulated data, and each set of samples should have the same dimensionality; (2) Configure the CycleGAN network structure and parameters to ensure the adversarial balance of the generator and discriminator; (3) Digital twin models for different fault types need to be trained separately.

3.4 | Advantages of the digital twin-assisted fault diagnosis system

Deep learning-based fault diagnosis requires a multitude of fault data, which is difficult to obtain. To solve the problem of insufficient training data, a digital twin-assisted fault system is proposed. There are three main benefits to the digital twin-assisted fault diagnosis system: (1) Only a small amount of measured data is required to generate simulated data with the same distribution and fault characteristics as the measured data. (2) The simulated data obtained utilizing the digital twin model can be directly used to train the classifier and provide high classification accuracy. (3) Higher accuracy and faster convergence can be obtained if the simulated data and the measured data are used together as the training samples.

4 | EXPERIMENTAL VERIFICATION

To verify the feasibility of the proposed digital twin-assisted fault diagnosis system, we build a test-rig to simulate the motion of the real robot joint. In this section, we will evaluate the simulated vibration signals generated by the CycleGAN-based digital twin model. The classification accuracy in different cases and the comparison with other fault diagnosis methods will also be presented in this section.

4.1 | Introduction of the test-rig

A RV reducer fault test-rig which can simulate the actual working conditions of the robot is built to collect the gear-fault data. The test-rig consists of a servo motor (MDMF152L1H6M), a RV reducer (BX-40E-121), an arm, a supporting platform, and several sensors, just as shown in Figure 9. The data is collected by Net0824 Data Acquisition Card. The sampling frequency is set to 6250 Hz, the sampling time is 2 min, and the data points for each sample are 750,000.

The details of the sensors are illustrated in Table 1. The three-component acceleration sensors are placed on the top of the RV reducer and collect the acceleration in three directions: *X*, *Y*, and *Z*. The temperature sensors are placed on the side of the RV reducer and servo motor to collect the temperature changes of the RV reducer and servo motor during operation. The RV reducer used in this experiment contains one sun wheel and two planetary wheels, as shown in Figure 10. And the parameters of the RV reducer are listed in Table 2.

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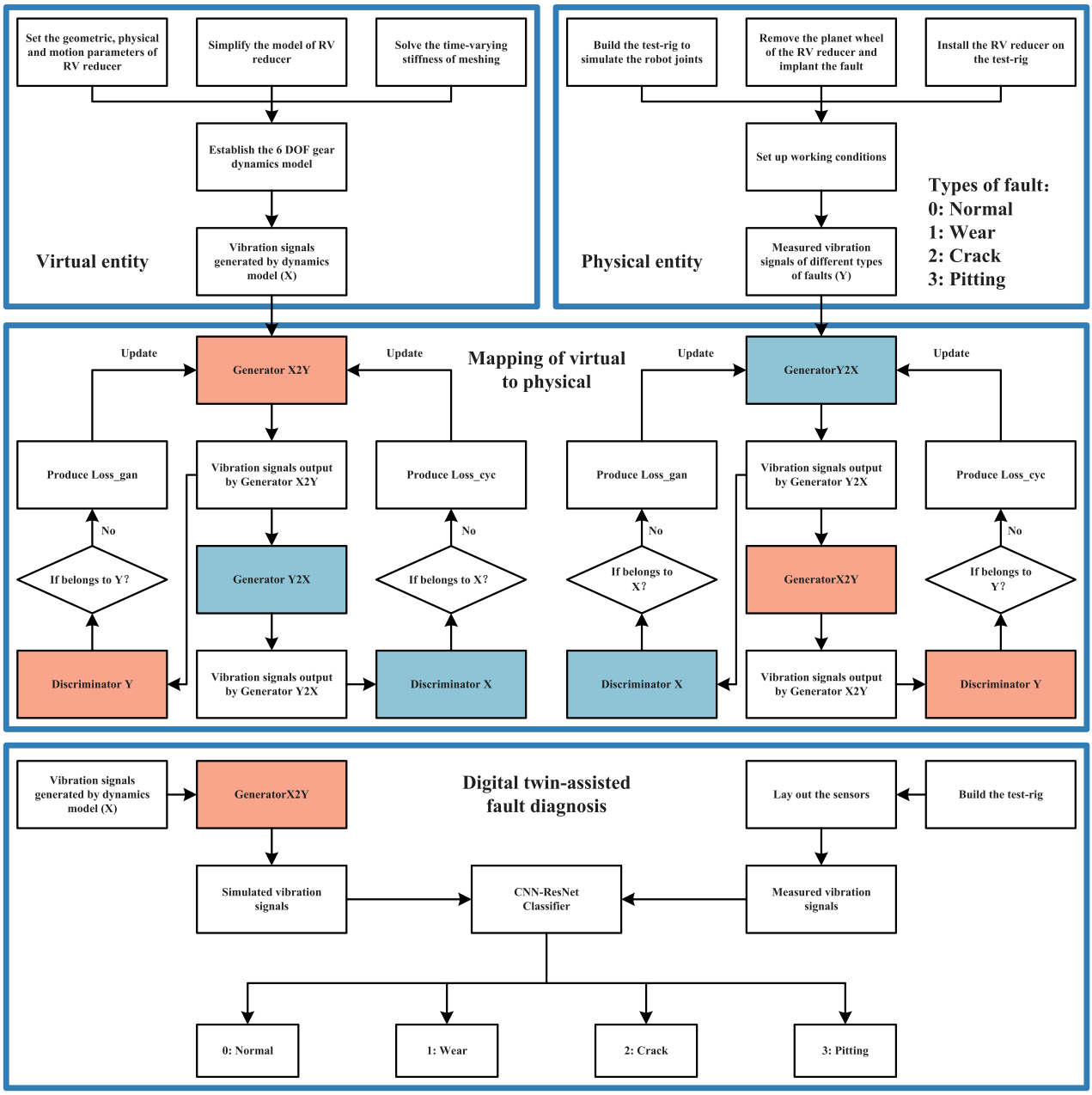


FIGURE 8 The mapping process of CycleGAN

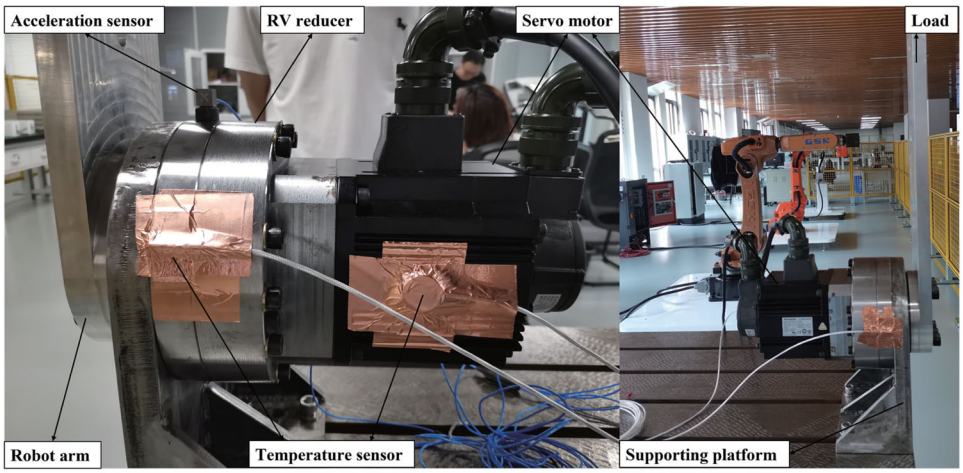


FIGURE 9 The test-rig of RV reducer

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The fault types of the experimental planetary gear are divided into three categories: wear, crack, and pitting. The description of the measured data is listed in Table 3, and the faulty planetary gears are shown in Figure 11. The following four types of measured vibration signals will be used as the Y-domain data for the subsequently introduced CycleGAN-based digital twin model.

4.2 | Virtual entity modeling

The virtual entity model based on gear dynamics can generate noiseless and faultless gear vibration signals, which are used as the domain X of the digital twin model. The parameters of the sun gear and the planetary gear are listed in Table 4. To facilitate verification of the validity of the dynamics model, the gear parameters used in the modeling are set to be the same as those of the gears in the test-rig.

Based on the established dynamic equation (6), the dynamics simulation of the faultless gear is carried out. According to the speed (45°/s) of the output shaft and the reduction ratio (121) of the RV reducer, the rotation frequency of the sun wheel can be calculated to be 25 Hz. And the meshing frequency (400 Hz) of the gear system in this work can be calculated from the rotation frequency and the number of teeth. Utilizing the parameters in Table 1, the acceleration dynamic response of the gear system can be solved utilizing the Runge-Kutta method. Figure 12a,b show the acceleration response and frequency spectrum, respectively. When the gear is faultless, the frequency spectrum primarily displays the meshing frequency and its multiples, as illustrated in Figure 12b. The meshing characteristics of the gears are well presented in these vibration signals generated by the virtual entity model. And these vibration signals will provide the X-domain date for the subsequently introduced CycleGAN-based digital twin model.

TABLE 1 Parameters of the sensors

Sensor	Type	Measuring range	Sensitivity
Acceleration sensor	EV450/ SN00064	−50 ~ 50 g	*X*: 99.60 mV/g
			Y: 98.37 mV/g
			Z: 99.28 mV/g
Temperature sensors	PT100	−72 ~ 200°C	37.037 MV/°C

4.3 | Model training and data generation

The CycleGAN-based digital twin model is trained with a small amount of real data to generate simulated vibration signals for fault diagnosis. To verify the generation capability of the digital twin model, all four types of vibration signals are generated. The X-domain data of the digital twin model is provided by the noiseless and faultless vibration signals generated by the virtual entity model. They are divided into groups of samples, where each group contains 1000 data points. And the Y-domain data of the digital twin model is provided by the measured data. To verify that the proposed digital twin model can solve the problem of insufficient training data, we use only a few measured data to generate simulated data for fault diagnosis. Each sample of Y-domain data contains 1000 data points, and only 100 samples of each type of Y-domain data are used. The digital twin model for each fault type needs to be trained separately. After four training sessions, the digital twin model for each fault type is saved for subsequent generation of simulated vibration signals. Figure 13 shows the time domain waveform and frequency spectrum of the measured and simulated data. It can be seen from Figure 13

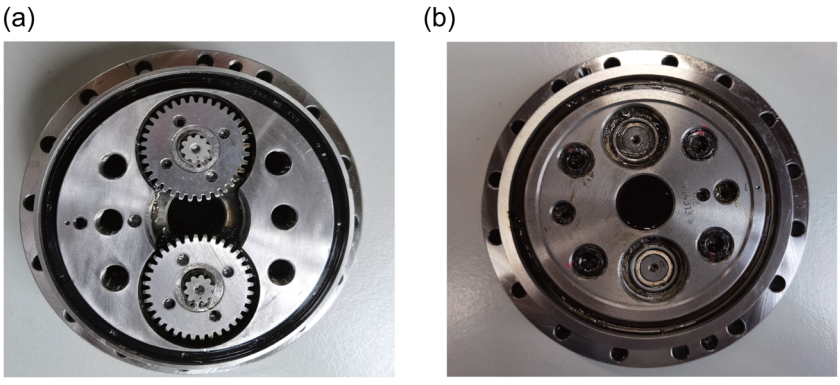


FIGURE 10 RV reducer used in the experiment: (a) Front side and (b) behind side

TABLE 2 Parameters of the RV reducer

Type	Reduction ratio	Rated output torque	Rated output speed	Allowable output speed	Weight
BX-40E-121	121	412 N m	15 r/min	70 r/min	9.3 kg

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that the simulated vibration signal generated by the digital twin model has a high degree of similarity with the measured vibration signal.

4.4 | Fault diagnosis analysis and discussion

First, we test the performance of the proposed CNN-ResNet classifier trained utilizing different amounts of measured data. The 600, 1200, 2400 and 4800 sets of training samples are input to the classifier respectively. The fault diagnosis accuracy is shown in Figure 14.

As shown in Figure 14, the fault diagnosis accuracy can reach almost 100% when the training samples are sufficient, such as the numbers of 2400 and 4800. However, the accuracy of the test set is quite low when the training samples are insufficient. When the training samples are 1200, the accuracy is about 39%. When the training samples are 600, the accuracy is only 32.5%. Therefore, it cannot provide a good performance of fault diagnosis by only utilizing the measured data in the case of insufficient training samples. Subsequently, we will introduce simulated data to improve the performance of fault diagnosis in the case of insufficient measured data.

To verify the validity of the simulated data generated by the digital twin model, we will test the simulated data utilizing the trained CNN-ResNet model. 4800 measured data are used to train the CNN-ResNet model, and each type of training sample includes 1200 measured data. After 100 epochs of training, we test the simulated data utilizing the trained CNN-ResNet model. The results show that the CNN-ResNet model trained with 4800 measured data can achieve almost 100% classification accuracy for the simulated data, and this result fully demonstrates that the simulated data generated by the digital twin model has the same distribution and fault characteristics as the measured data.

Next, we will investigate the effect of different numbers of training samples on the performance of fault diagnosis. Nine cases containing different amounts of the measured and simulated data are set, and the fault diagnosis accuracy for each case is listed in Table 5. The classification accuracy during model learning for these nine cases is shown in Figure 15.

Based on the results shown above, it can be seen that sufficient simulated vibration signals can improve the accuracy of fault diagnosis. As the training sample increases, the model converges faster. And when the measured data is used as the training samples along with the simulated data, the convergence speed of the model

TABLE 3 Description of the measured data

Label	Fault type	Load (kg)	Rotational speed	Data points
0	Normal	24	45°/s	7,500,000
1	Wear	24	45°/s	7,500,000
2	Crack	24	45°/s	7,500,000
3	Pitting	24	45°/s	7,500,000

TABLE 4 Parameters of the sun gear and the planetary gear

Parameters	Sun gear	Planetary gear
Number of teeth	12	36
Modulus (mm)	1.5	1.5
Pressure angle (°)	20	20
Poisson's ratio	0.3	0.3
Elastic modulus (GPa)	206.8	206.8
Tooth width (mm)	10	10
Density (kg m^{-3})	7800	7800
Mass (kg)	0.06	0.28

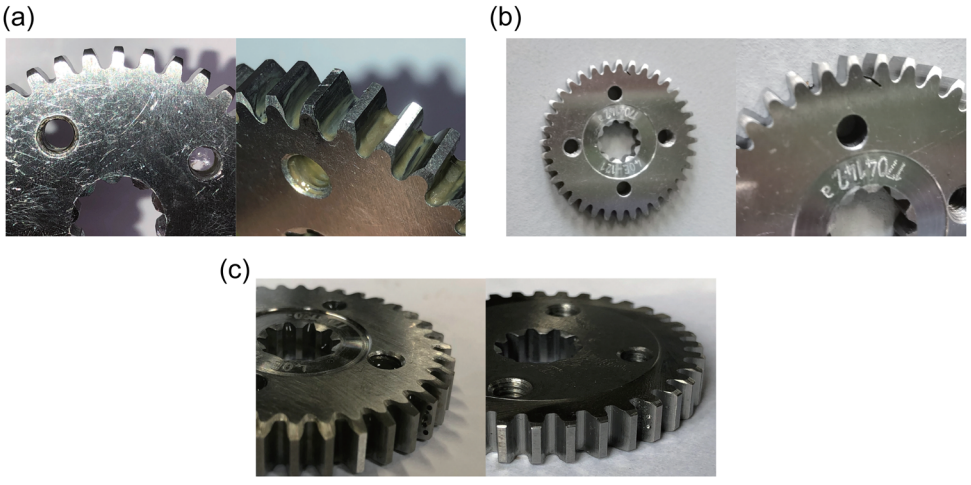


FIGURE 11 Fault type of planetary gear: (a) Wear, (b) crack, and (c) pitting

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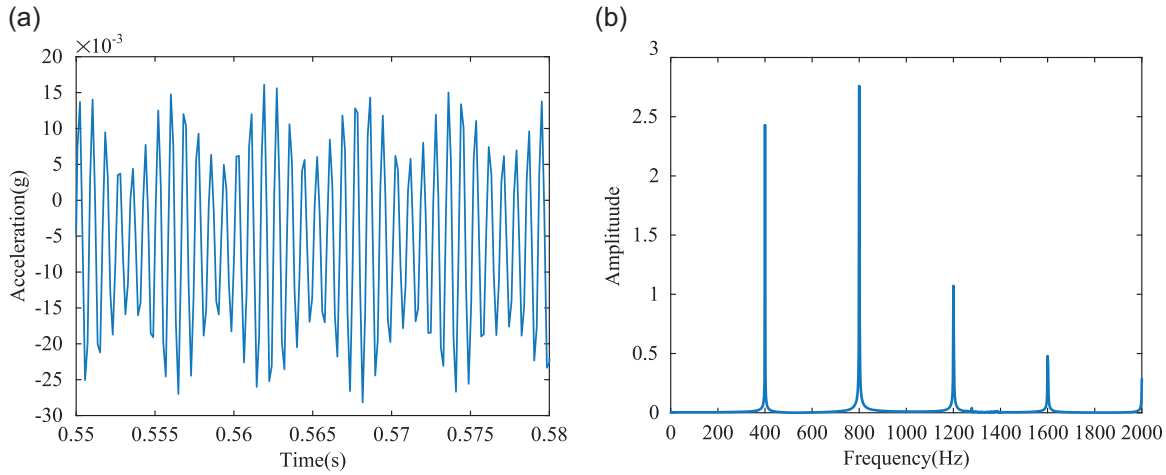


FIGURE 12 Dynamics simulation of faultless gear: (a) Acceleration response and (b) frequency spectrum

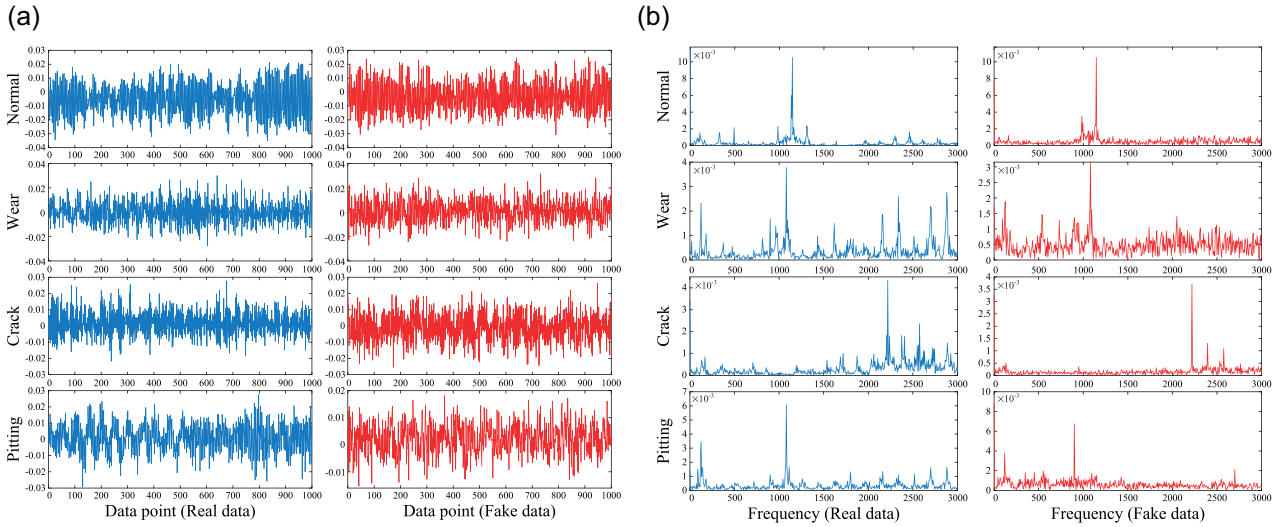


FIGURE 13 Comparison of the real data with the fake data: (a) Time domain waveform and (b) frequency spectrum

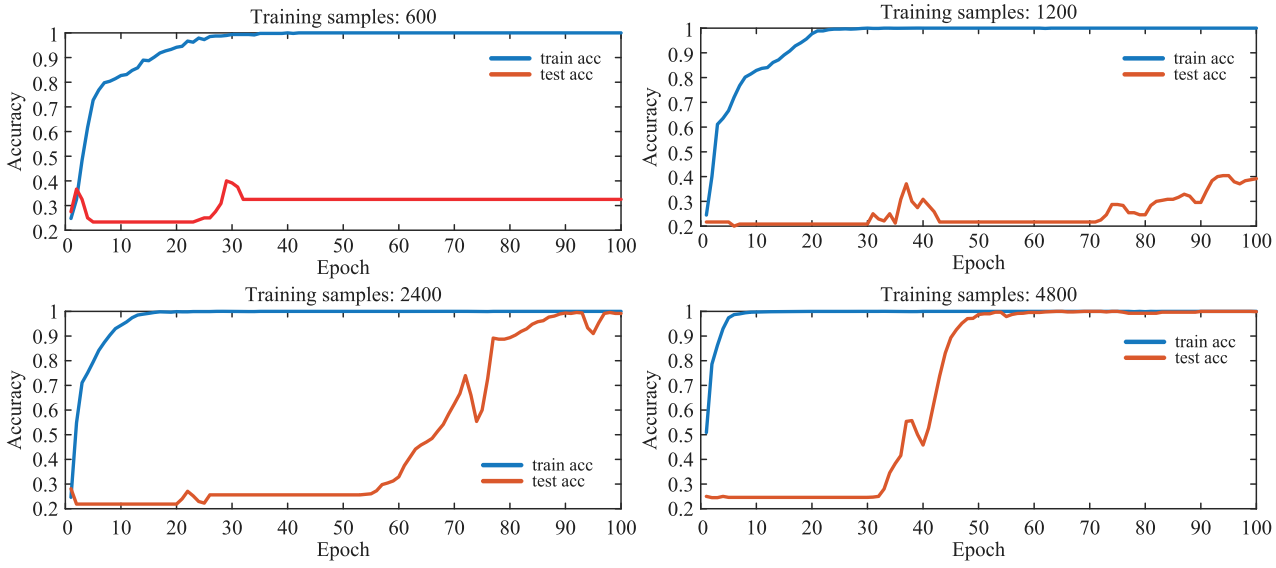


FIGURE 14 Fault diagnosis accuracy with different training samples

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TABLE 5 Training sample settings containing different measured data and simulated data

Case	1	2	3	4	5	6	7	8	9
Real samples	400	0	0	0	0	400	400	400	400
Fake samples	0	400	600	800	1200	400	600	800	1200
Accuracy (%)	32.50	82.57	96.88	98.81	98.86	98.51	99.24	100	100

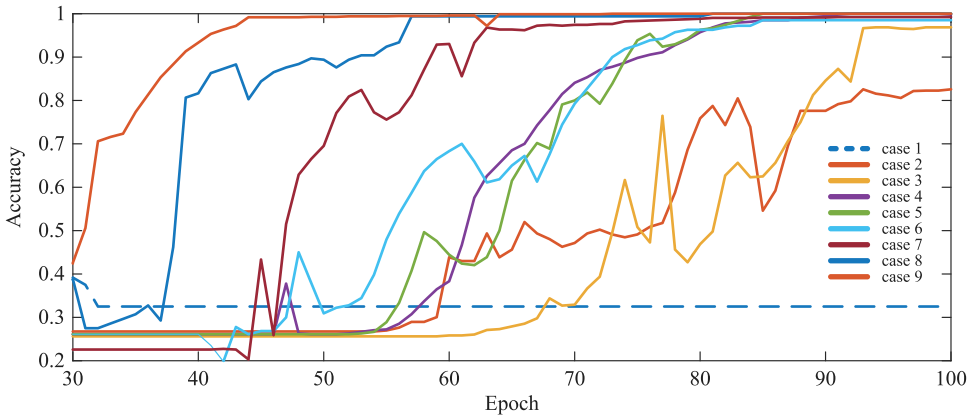


FIGURE 15 Classification accuracy in different cases

will be faster, and the fault diagnosis accuracy will be further improved. All of the above-simulated signals are obtained by utilizing only 400 measured data to train the CycleGAN-based digital twin model. And utilizing 1200 simulated data as training samples, the CNN-ResNet model can provide 98.86% fault diagnosis accuracy. However, if the CNN-ResNet model is trained utilizing 400 measured data as training samples directly, the model provides only 32.5% of the fault diagnosis accuracy. Therefore, the proposed digital twin-assisted fault diagnosis system can improve the fault diagnosis accuracy under the condition of a small amount of measured data and effectively solve the problem of insufficient training data.

4.5 | Comparison with other fault diagnosis methods

To further highlight the superiority of the digital twin-assisted fault diagnosis system, we compare it with several of the better-performing fault diagnosis methods. Other mainstream fault diagnosis methods that can solve the insufficient training data problem are ML1D-GAN (Guo et al., 2020), ACGAN (Shao et al., 2019b), CACGAN (Dixit et al., 2021), transfer learning (Shao et al., 2019a), and so on. To compare the ability of these methods to solve the problem of insufficient data, we use as little measured data as possible while ensuring fault accuracy. The comparison of the different fault diagnosis methods is presented in Table 6. It should be noted that the transfer learning-based fault diagnosis methods require a large amount of data to pre-train the model. Table 6 only records the measured data required for the secondary training of the transfer learning model, and the measured data required for the pre-training is not recorded in this experiment.

TABLE 6 The comparison of the different fault diagnosis methods

Method	Measured data	Accuracy (%)
ML1D-GAN	400	82.33
	800	97.79
ACGAN	400	86.45
	800	96.41
CACGAN	400	94.92
	800	97.18
Transfer Learning	400	48.57
	1600	99.37
Proposed	400	98.86

As shown in Table 6, the classification accuracy of the proposed method (98.83%) is higher than that of CACGAN (94.92%), ACGAN (86.45%), ML1D-GAN (82.33%) and Transfer learning (48.57%), when utilizing 400 sets of measured data. When utilizing 800 sets of measured data, the fault diagnosis accuracy of ML1D-GAN (97.79%) is higher than that of CACGAN (97.18%) and ACGAN (96.41%), but still does not exceed our proposed method. This means that the method proposed in this paper performs better. It is worth noting that the Transfer Learning-based fault diagnosis method presents the highest accuracy (99.37%), but it requires more measured data than

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other methods. And it also requires a large amount of data to pre-train the model. Therefore, it is not as economical as other methods. The above results demonstrate that the digital twin-assisted fault diagnosis system proposed in this paper can obtain higher fault diagnosis accuracy with fewer measured data.

5 | CONCLUSION

In this paper, a digital twin-assisted fault diagnosis system is proposed for solving the problem of insufficient training data. The main contributions are as follows.

1. There are a multitude of problems in traditional manual building construction technology, which have become the shortcomings of the new intelligent building construction technology. The development of construction robots with a high automation level has become an inevitable trend in the new era. To ensure the normal operation and accurate work of construction robots, a digital twin-assisted fault diagnosis system is proposed in this work for the robot joint (RV reducer). The system can effectively diagnose faults in robot joints with insufficient data.
2. A CycleGAN-based digital twin model is proposed to generate different types of simulated data for fault diagnosis. Based on the features of CycleGAN, the model can map the X-domain data to the Y-domain data. The noiseless and faultless vibration signals generated by gear dynamics are the X-domain data of this digital twin model, and the measured data of various fault types are the Y-domain data. The digital twin model requires only a small number of measured data as the training samples to generate the simulated data, which has the same distribution and fault characteristics as the measured data.
3. To verify the feasibility of the digital twin-assisted fault diagnosis system, we establish a test-rig simulating robot joint conditions and propose a CNN-ResNet classifier. First, we train the CNN-ResNet model by utilizing different amounts of measured data. The results show that the fault diagnosis accuracy and convergence speed of the model decrease as the training sample decreases. Then, only 400 samples of measured data are used to generate the simulated vibration signals, which can be directly used for fault diagnosis. The results show that the simulated vibration signals can provide 98.86% accuracy, which can demonstrate that the proposed method can solve the problem of insufficient training data. Finally, the proposed method is compared with other mainstream methods. The results demonstrate that the proposed method can obtain higher fault diagnosis accuracy with fewer measured data.

ACKNOWLEDGMENTS

The authors would like to thank the financial supports from National Science Foundation of China (Nos. 52075348, 52275119), High-level innovation and entrepreneurship team (No. XLYC2008016), Liaoning Revitalization Talents Program (No. XLYC2007031), and Natural Science Foundation of Liaoning (No. 2021-KF-12-09). The authors would also like to thank Professor Yaguo Lei and graduate student Ping He of Xi'an Jiaotong University for providing support on the test-rig and the data.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Song, Z., Shi, H., Bai, X. & Li, G. (2023) Digital twin-assisted fault diagnosis system for robot joints with insufficient data. *Journal of Field Robotics*, 40, 258–271. https://doi.org/10.1002/rob.22127

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